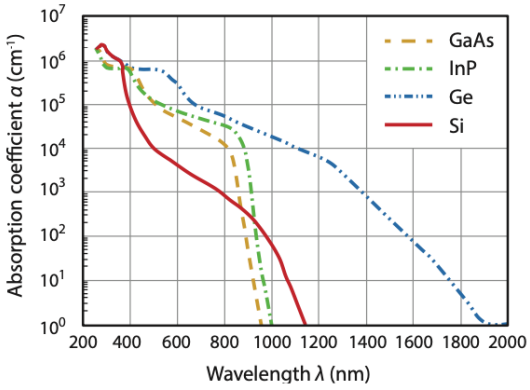
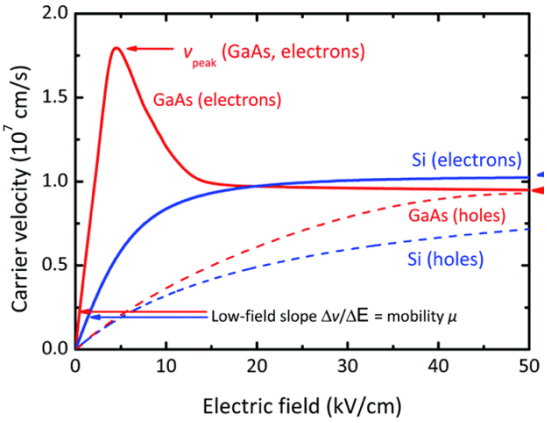
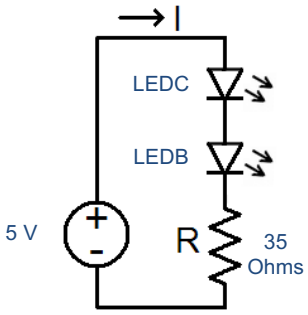
1) A GaAs PIN photodiode has an intrinsic region of thickness 0.1 mm. Determine the transit time for the detector and the required reverse-bias voltage. For this photodiodes, determine the efficiency for 800 nm light (take refractive index as 3.6 and ξ=1).

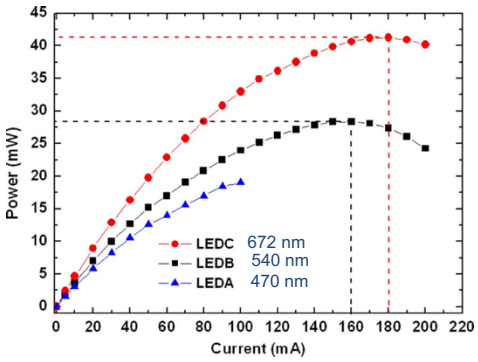
2) Describe a method to determine the optimal load resistance connected across a solar cell for the best fill factor.



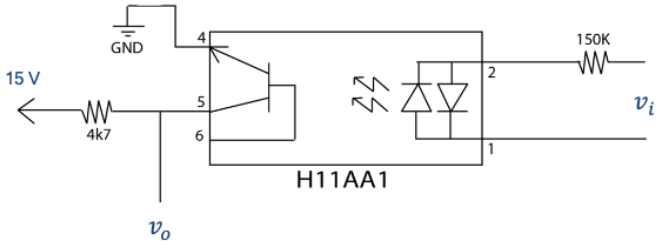


3) For the given LED circuit, find the luminous flux of each LED. Both diodes have a built-in voltage of 1 V. The PI curves of the diodes are as shown.





4) In the circuit below, the input is a sinusoidal signal. Sketch the input and output voltage signals together on the same figure. What is the function of the circuit?

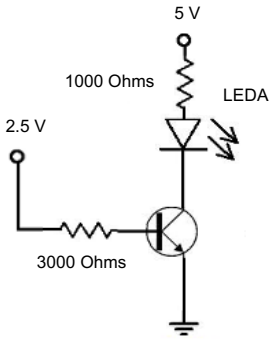


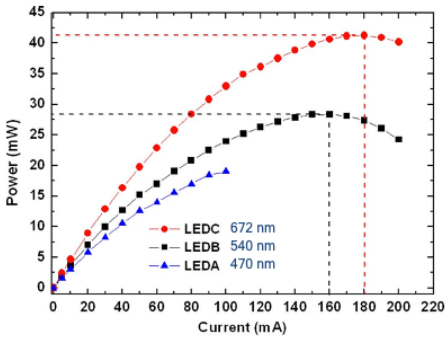
5) Find the peak wavelength, spectral width and escape cone angle of an LED made of $Al_{0.25}Ga_{0.75}As$ at room temperature. Take $\Delta E = 3\ k_B T$. Show all work.

6) What are the two most important reasons why efficient light emission from p-n junctions occurs only in forward bias? Explain.

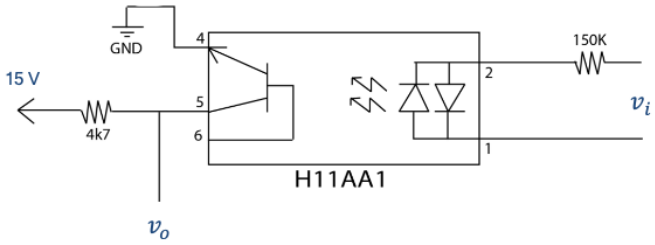
7) Discuss two methods to solve the problem of absorption of generated photons inside an LED.

8) Find the luminous flux of the LED in the circuit below. LEDA has a built-in voltage of 1 V and the transistor has β=120. Use the given PI curves.





9) Keeping the H11AA1, re-design the circuit below to detect the peaks of the input? Show all details and explain your design.



10) Explain the effects of leakage current on the performance of an LED. How can we reduce these effects?

11) Suggest two ways for increasing the modulation bandwidth of an LED. Explain.